



ULUSLARARASI KIBRIS ÜNİVERSİTESİ  
CYPRUS INTERNATIONAL UNIVERSITY

**CYPRUS INTERNATIONAL UNIVERSITY**

**FACULTY of ENGINEERING**

**DATABASE MANAGEMENT SYSTEMS AND  
PROGRAMMING II**

**STRAWBERRY FARMING CORP.**

**Students and Roles ;**

- **Fatmatül Zehra Taş (22208075):** DDL (table creation, constraints), DML (insert, update, delete operations), database connection (Npgsql), SQL queries, and User Interface (UI) development
- **Hediye Sultan Bozkurt (22319266):** Report writing, PL/pgSQL (procedures, functions, triggers), and advanced SQL queries
- **Zeynep Özlüer (22315460):** ER Diagram (ERD) design, database modeling, and contribution to SQL queries.
- **Emre Özel (22315566):** User Interface (UI) design (WinForms), application layout, and database integration

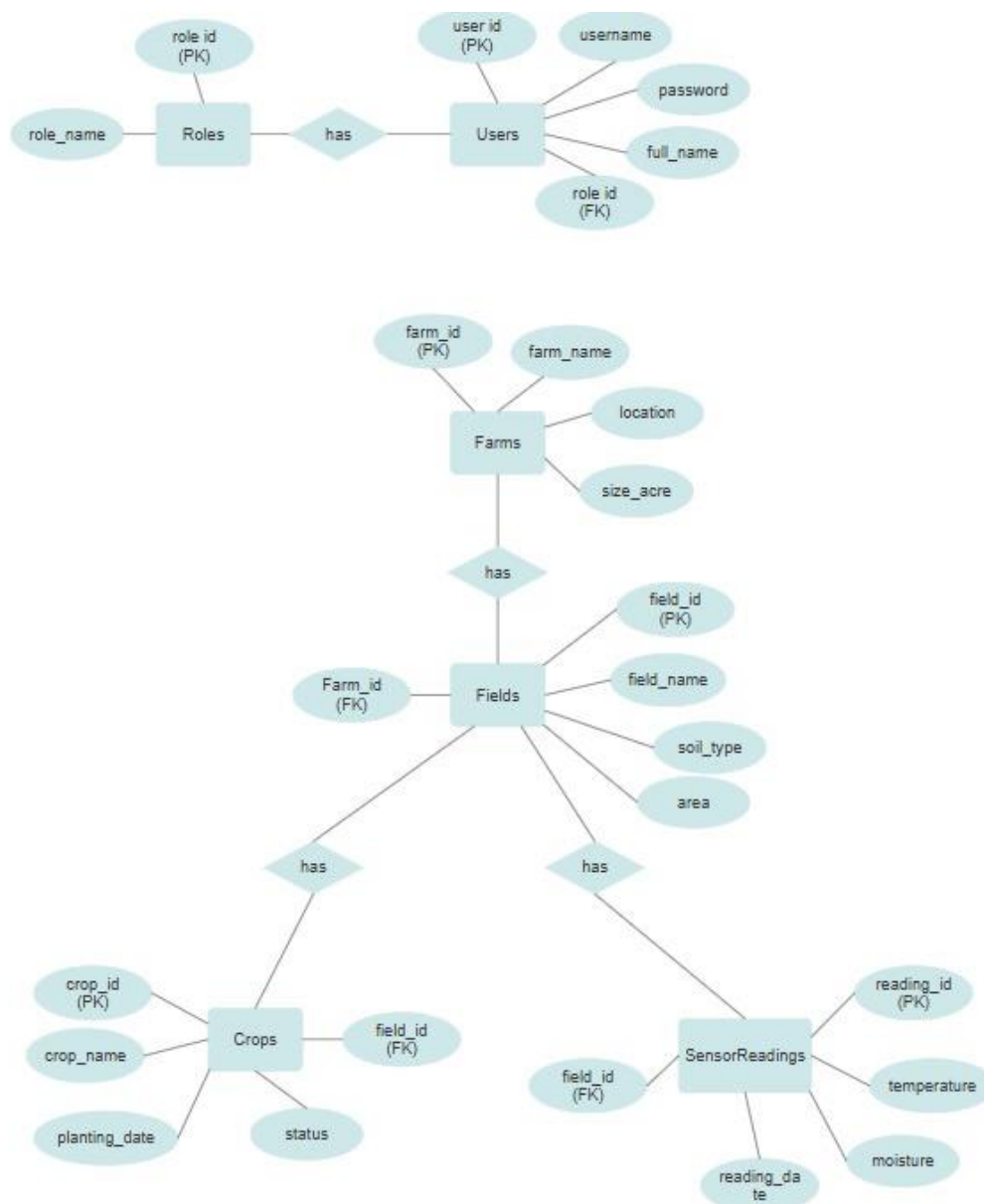
**Instructor: *PROF. DR. MELİKE ŞAH DİREKOĞLU***

## Introduction

This project presents a Smart Farming Database Management System designed to manage farms, fields, crops, and sensor data such as temperature and moisture. The system allows users to store, update, and analyze data through a user-friendly interface.

It assumes that each farm contains multiple fields, and each field has crops and sensor readings. The system helps monitor conditions and detect risks, supporting better decision-making in agriculture.

### 1) ER DIAGRAM



## 2) Data Definition Language Details (DDL)

### Tables

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```
1  |-- ===== TABLES =====
2
3  CREATE TABLE Roles (
4      role_id SERIAL PRIMARY KEY,
5      role_name TEXT NOT NULL
6  );
7
8  CREATE TABLE Users (
9      user_id SERIAL PRIMARY KEY,
10     username TEXT NOT NULL,
11     password TEXT NOT NULL,
12     full_name TEXT NOT NULL,
13     role_id INT,
14     FOREIGN KEY (role_id)
15         REFERENCES Roles(role_id)
16         ON DELETE SET NULL
17 );
18
19 CREATE TABLE Farms (
20     farm_id SERIAL PRIMARY KEY,
21     farm_name TEXT NOT NULL,
22     location TEXT NOT NULL,
23     size_acre NUMERIC
24 );
```

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```
25
26 CREATE TABLE Fields (
27     field_id SERIAL PRIMARY KEY,
28     farm_id INT,
29     field_name TEXT NOT NULL,
30     soil_type TEXT,
31     area NUMERIC,
32     FOREIGN KEY (farm_id)
33         REFERENCES Farms(farm_id)
34         ON DELETE CASCADE
35 );
36
37 CREATE TABLE Crops (
38     crop_id SERIAL PRIMARY KEY,
39     field_id INT,
40     crop_name TEXT NOT NULL,
41     planting_date DATE,
42     status TEXT,
43     FOREIGN KEY (field_id)
44         REFERENCES Fields(field_id)
45         ON DELETE CASCADE
46 );
47
48 CREATE TABLE SensorReadings (
49     reading_id SERIAL PRIMARY KEY,
50     field_id INT,
51     temperature INT,
52     moisture INT,
53     reading_date TIMESTAMP,
54     FOREIGN KEY (field_id)
55         REFERENCES Fields(field_id)
56         ON DELETE CASCADE
57 );
```

Connected (12 queries)

Run Explain Analyze

- croops
- farms
- fields
- roles
- sensorreadings
- users

Values

UntitledSavePrimaryActive▼neondb⌂⚙

```
56 -- CREATE SCHEMA
57 };
58
59 -- ===== DATA =====
60
61 INSERT INTO Roles (role_name) VALUES
62 ('Admin'),
63 ('Technician'),
64 ('Farmer');
65
66 INSERT INTO Users (username, password, full_name, role_id) VALUES
67 ('admin', '1234', 'System Admin', 1),
68 ('tech1', '1234', 'Hydroponic Technician', 2),
69 ('farmer1', '1234', 'Strawberry Grower', 3);
70
71 INSERT INTO Farms (farm_name, location, size_acre) VALUES
72 ('Hydro Strawberry Farm', 'Micosia', 12),
73 ('Smart Berry Lab', 'Kyrenia', 8);
74
75 INSERT INTO Fields (farm_id, field_name, soil_type, area) VALUES
76 (1, 'Greenhouse A', 'Hydroponic', 3),
77 (1, 'Greenhouse B', 'Hydroponic', 4),
78 (2, 'Vertical Tower 1', 'Hydroponic', 2);
79
80 INSERT INTO Crops (field_id, crop_name, planting_date, status) VALUES
81 (1, 'Strawberry Albion', '2026-03-01', 'Growing'),
82 (2, 'Strawberry Festival', '2026-02-15', 'Ready'),
83 (3, 'Strawberry San Andreas', '2026-03-10', 'Planted');
84
85 INSERT INTO SensorReadings (field_id, temperature, moisture, reading_date) VALUES
86 (1, 22, 70, '2026-05-06 10:00:00'),
87 (2, 24, 65, '2026-05-06 10:05:00'),
88 (3, 21, 75, '2026-05-06 10:10:00');
```

Ready to connect

Run

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1 INSERT INTO farms(farm\_name) VALUES ('Hydrd Strawberry Farm');

Connected {1 query}

Run | Explain | Analyze

ERROR: Farm name already exists! (SQLSTATE P0001)

Fix with AI ⚡

| # | field_id | farm_id | field_name       | soil_type  | area |
|---|----------|---------|------------------|------------|------|
| 1 | 1        | 1       | Greenhouse A     | Hydroponic | 3    |
| 2 | 2        | 1       | Greenhouse B     | Hydroponic | 4    |
| 3 | 3        | 2       | Vertical Tower 1 | Hydroponic | 2    |

| # | crop_id | field_id | crop_name              | planting_date | status  |
|---|---------|----------|------------------------|---------------|---------|
| 1 | 1       | 1        | Strawberry Albion      | 2026-03-01    | Growing |
| 2 | 2       | 2        | Strawberry Festival    | 2026-02-15    | Ready   |
| 3 | 3       | 3        | Strawberry San Andreas | 2026-03-10    | Planted |

| # | farm_id | farm_name             | location | size_acre |
|---|---------|-----------------------|----------|-----------|
| 1 | 1       | Hydro Strawberry Farm | Nicosia  | 12        |
| 2 | 2       | Smart Berry Lab       | Kyrenia  | 8         |

| # | role_id | role_name  |
|---|---------|------------|
| 1 | 1       | Admin      |
| 2 | 2       | Technician |
| 3 | 3       | Farmer     |

| # | reading_id | field_id | temperature | moisture | reading_date        |
|---|------------|----------|-------------|----------|---------------------|
| 1 | 1          | 1        | 22          | 70       | 2026-05-06 10:00:00 |
| 2 | 2          | 2        | 24          | 65       | 2026-05-06 10:05:00 |
| 3 | 3          | 3        | 21          | 75       | 2026-05-06 10:10:00 |

| # | user_id | username | password | full_name             | role_id |
|---|---------|----------|----------|-----------------------|---------|
| 1 | 1       | admin    | 1234     | System Admin          | 1       |
| 2 | 2       | tech1    | 1234     | Hydroponic Technician | 2       |
| 3 | 3       | farmer1  | 1234     | Strawberry Grower     | 3       |

### 3) Data Manipulation Language (DML)

1.Query: This query finds fields whose average temperature is higher than the overall average temperature of all sensor readings.

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Primary

Active

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1

2

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9

SELECT f.field\_name,

AVG(s.temperature) AS avg\_temp

FROM sensorreadings s

JOIN fields f ON s.field\_id = f.field\_id

GROUP BY f.field\_name

HAVING AVG(s.temperature) > (

SELECT AVG(temperature)

FROM sensorreadings

);

Connected (1 query)

Run

Explain

Analyze

215ms1 row

| # | field_name   | avg_temp           |
|---|--------------|--------------------|
| 1 | Greenhouse B | 24.000000000000000 |

StatisticsForm

STRAWBERRY FARMING CORP

Welcome! Have a nice day

What you want to see?

Above Avg Temperature

Crops Per Farm

Min Max Temperature

Highest Moisture Field

Latest Sensor Data

Empty Fields

BACK

STATISTIC TABLE VIEW

|   | field_name   | avg_temp           |
|---|--------------|--------------------|
| ▶ | Greenhouse B | 24,000000000000... |
| * |              |                    |

2.Query: This query calculates the total number of crops in each farm by joining farms, fields, and crops tables.

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1SELECT fa.farm\_name, COUNT(c.crop\_id) AS total\_crops

2FROM farms fa

3JOIN fields f ON fa.farm\_id = f.farm\_id

4JOIN crops c ON f.field\_id = c.field\_id

5GROUP BY fa.farm\_name

6ORDER BY total\_crops DESC;

✓ Connected (1 query)

▶ Run

Explain

Analyze

230ms2 rc

| # | farm_name             | total_crops |
|---|-----------------------|-------------|
| 1 | Smart Berry Lab       | 2           |
| 2 | Hydro Strawberry Farm | 1           |

StatisticsForm

STRAWBERRY FARMING CORPWelcome! Have a nice day

What you want to see?

Above Avg Temperature

Crops Per Farm

Min Max Temperature

Highest Moisture Field

Latest Sensor Data

Empty Fields

BACK

STATISTIC TABLE VIEW

|   | farm_name         | total_crops |
|---|-------------------|-------------|
| ▶ | Hydro Strawber... | 2           |
|   | Smart Berry Lab   | 1           |
| * |                   |             |

3.Query: This query shows the minimum and maximum temperature recorded in each field.

Untitled

Save

Primary Active

neondb

```
1 SELECT f.field_name,
2       MAX(s.temperature) AS max_temp,
3       MIN(s.temperature) AS min_temp
4 FROM sensorreadings s
5 JOIN fields f ON s.field_id = f.field_id
6 GROUP BY f.field_name;
```

Connected (1 query)

Run

Explain

Analyze

195ms

3 rows

| # | field_name       | max_temp | min_temp |
|---|------------------|----------|----------|
| 1 | Vertical Tower 1 | 23       | 23       |
| 2 | Greenhouse B     | 22       | 22       |
| 3 | Greenhouse A     | 22       | 22       |

StatisticsForm

STRAWBERRY FARMING CORP

Welcome! Have a nice day

What you want to see?

Above Avg Temperature

Crops Per Farm

Min Max Temperature

Low Moisture Analysis

Latest Sensor Data

Empty Fields

BACK

STATISTIC TABLE VIEW

|   | field_name       | max_temp | min_temp |
|---|------------------|----------|----------|
| ▶ | Vertical Tower 1 | 23       | 23       |
|   | Greenhouse B     | 22       | 22       |
|   | Greenhouse A     | 22       | 22       |
| * |                  |          |          |



4.Query: This query finds the field with the highest moisture value by calculating the maximum moisture per field and returning the top result.

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```
1 SELECT f.field_name,
2       MAX(s.moisture) AS highest_moisture
3 FROM "fields" f
4 JOIN "sensorreadings" s
5 ON f.field_id = s.field_id
6 GROUP BY f.field_name
7 ORDER BY highest_moisture DESC
8 LIMIT 1;
```

✓ Connected (1 query)

▶ Run

Explain

Analyze

236ms1 row

| # | field_name   | highest_moisture |
|---|--------------|------------------|
| 1 | Greenhouse A | 70               |

StatisticsForm

STRAWBERRY FARMING CORP

Welcome! Have a nice day

What you want to see?

Above Avg Temperature

Crops Per Farm

Min Max Temperature

**Highest Moisture Field**

Latest Sensor Data

Empty Fields

BACK

STATISTIC TABLE VIEW

|   | field_name   | highest_moisture |
|---|--------------|------------------|
| ▶ | Greenhouse A | 70               |
| * |              |                  |

5.Query: This query retrieves the most recent temperature reading for each field by selecting records with the latest reading date from the sensorreadings table.

Untitled

Save

Primary Active

neondb

```
1 SELECT f.field_name, s.temperature, s.reading_date
2 FROM sensorreadings s
3 JOIN fields f ON s.field_id = f.field_id
4 WHERE s.reading_date = (
5     SELECT MAX(s2.reading_date)
6     FROM sensorreadings s2
7     WHERE s2.field_id = s.field_id
8 );
```

Connected (1 query)

Run

Explain

Analyze

207ms

2 rows

| # | field_name   | temperature | reading_date        |
|---|--------------|-------------|---------------------|
| 1 | Greenhouse A | 22          | 2026-05-06 10:00:00 |
| 2 | Greenhouse B | 24          | 2026-05-06 10:05:00 |

StatisticsForm

 STRAWBERRY FARMING CORP

Welcome! Have a nice day

What you want to see?

Above Avg Temperature

Crops Per Farm

Min Max Temperature

Highest Moisture Field

**Latest Sensor Data**

Empty Fields

BACK

STATISTIC TABLE VIEW

|   | field_name   | temperature | reading_date    |
|---|--------------|-------------|-----------------|
| ▶ | Greenhouse A | 22          | 6.05.2026 10:00 |
|   | Greenhouse B | 24          | 6.05.2026 10:05 |
| * |              |             |                 |

6.Query: This query lists all fields that do not currently contain any crops.

Untitled

Save

Primary Active

neondb

1

2

3

4

SELECT f.field\_name

FROM fields f

LEFT JOIN crops c ON f.field\_id = c.field\_id

WHERE c.crop\_id IS NULL;

Connected (1 query)

Run

Explain

Analyze

246ms

No result

Statement executed successfully

StatisticsForm

STRAWBERRY FARMING CORP

Welcome! Have a nice day

What you want to see?

Above Avg Temperature

Crops Per Farm

Min Max Temperature

Low Moisture Analysis

Latest Sensor Data

Empty Fields

BACK

STATISTIC TABLE VIEW

|   | field_name |
|---|------------|
| * |            |

Update (extra) Query: This query updates the status of a specific crop to "Harvested" based on its ID. It simulates a real-world scenario where a crop has completed its growth cycle.

Untitled

Save

Primary 

Active

neondb

1

2

3

4

5

6

UPDATE crops

SET status = 'Harvested'

WHERE crop\_id = 1;

SELECT crop\_id, crop\_name, status

FROM crops

WHERE crop\_id = 1;

Connected (2 queries)

Run

Explain

Analyze

189ms

1 row

1: UPDATE 1

2: SELECT 1

| # | crop_id | crop_name         | status    |
|---|---------|-------------------|-----------|
| 1 | 1       | Strawberry Albion | Harvested |

Delete(extra) Query: This query deletes a specific sensor reading from the system using its ID. It is useful for removing incorrect or outdated data.

Untitled

Save

Primary 

Active

neondb

1

2

3

DELETE FROM sensorreadings

WHERE reading\_id = 3;

SELECT \* FROM sensorreadings;

Connected (2 queries)

Run

Explain

Analyze

224ms

2 rows

1: DELETE 1

2: SELECT 2

| # | reading_id | field_id | temperature | moisture | reading_date        |
|---|------------|----------|-------------|----------|---------------------|
| 1 | 1          | 1        | 22          | 70       | 2026-05-06 10:00:00 |
| 2 | 2          | 2        | 24          | 65       | 2026-05-06 10:05:00 |

3) PL/SQL QUERIES

1, This procedure inserts a new crop into the database while preventing duplicate crop entries in the same field.

Untitled

Save

Primary Active

neondb

```
1 CREATE OR REPLACE PROCEDURE add_crop(  
2     p_field_id INT,  
3     p_crop_name TEXT,  
4     p_date DATE  
5 )  
6 LANGUAGE plpgsql  
7 AS $$  
8 BEGIN  
9     IF NOT EXISTS (  
10         SELECT 1 FROM crops  
11         WHERE field_id = p_field_id AND crop_name = p_crop_name  
12     ) THEN  
13         INSERT INTO crops(field_id, crop_name, planting_date, status)  
14         VALUES (p_field_id, p_crop_name, p_date, 'New');  
15     ELSE  
16         RAISE NOTICE 'Crop already exists in this field';  
17     END IF;  
18 END;  
19 $$;  
20 CALL add_crop(1, 'StrawberryX', '2026-05-14');
```

✓ Connected (2 queries)

▶ Run

Explain

Analyze

275ms

No result

1: CREATE    2: CALL

Statement executed successfully

Untitled

Save

Primary Active

neondb

```
1 SELECT crop_name, field_id FROM crops;
```

✓ Connected (1 query)

▶ Run

Explain

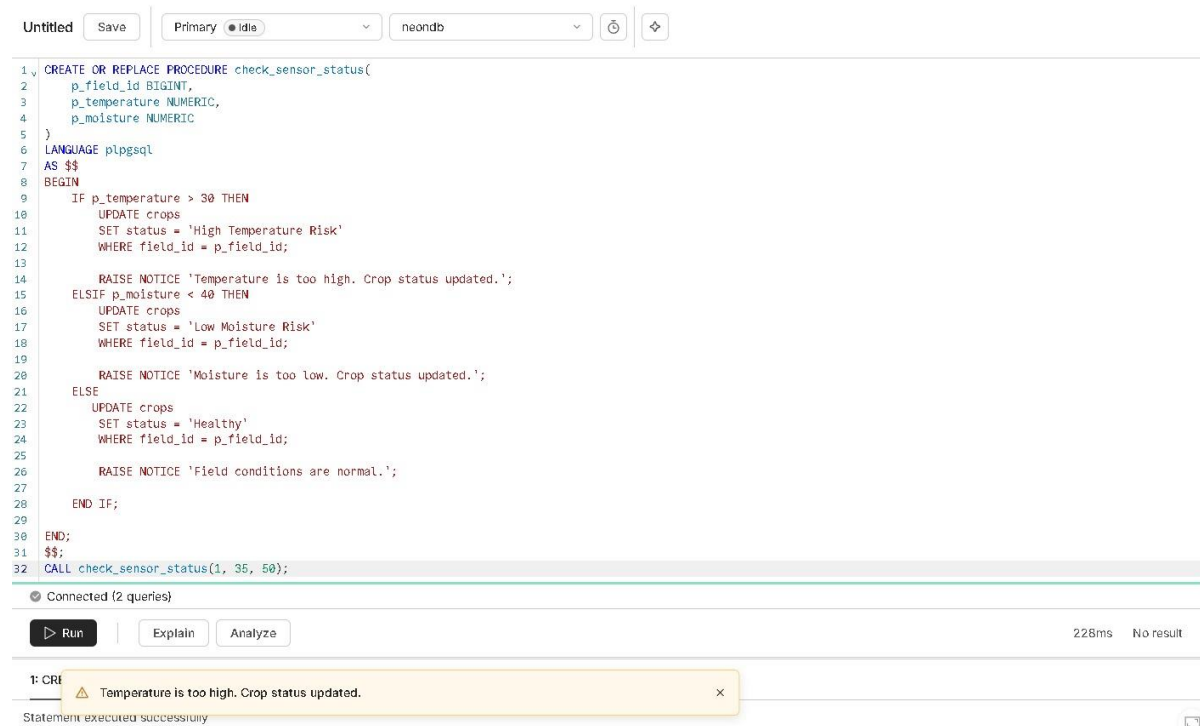
Analyze

197ms

1 row

| # | crop_name           | field_id |
|---|---------------------|----------|
| 1 | Strawberry Festival | 2        |

2, This procedure automatically updates a field's crop status in the database to a specific risk level or "Healthy" by evaluating incoming temperature and moisture sensor data.



The screenshot shows a SQL IDE interface with a code editor, a toolbar, and a results pane. The code editor contains a PL/SQL procedure named `check_sensor_status` that takes `p_field_id`, `p_temperature`, and `p_moisture` as parameters. The procedure uses conditional logic to update the status of a crop based on temperature and moisture levels. The results pane shows the execution of the procedure with the following output:

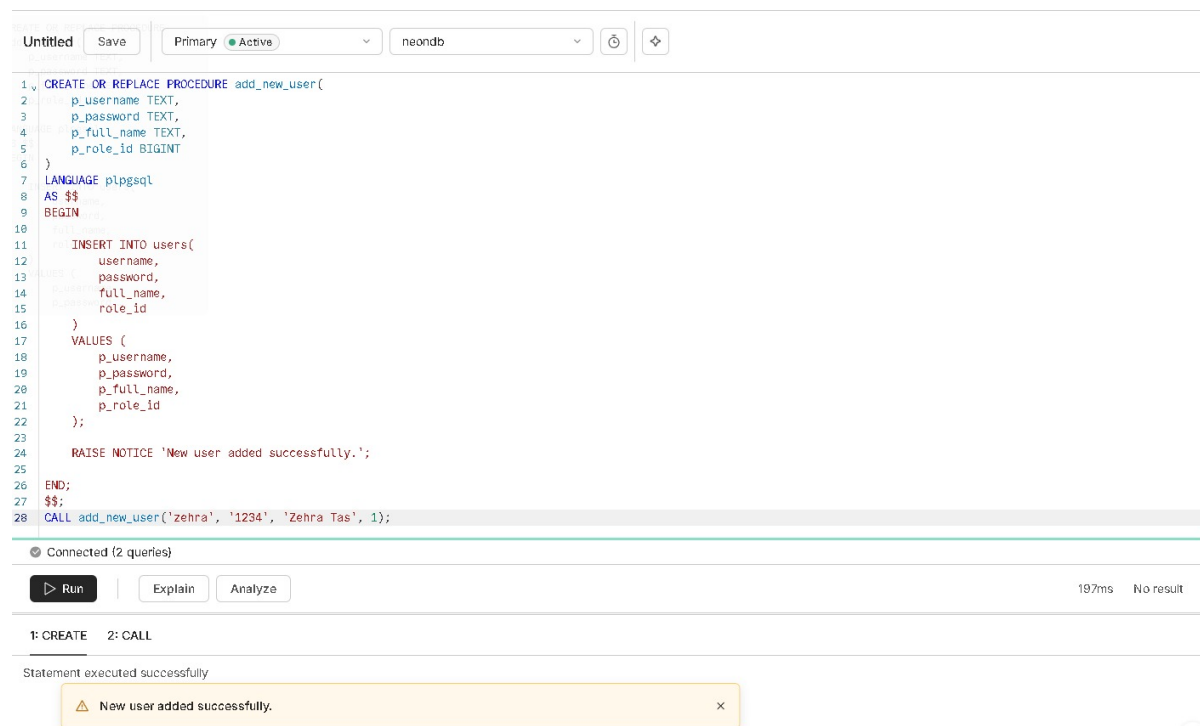
```
1 CREATE OR REPLACE PROCEDURE check_sensor_status(  
2   p_field_id BIGINT,  
3   p_temperature NUMERIC,  
4   p_moisture NUMERIC  
5 )  
6 LANGUAGE plpgsql  
7 AS $$  
8 BEGIN  
9   IF p_temperature > 30 THEN  
10    UPDATE crops  
11    SET status = 'High Temperature Risk'  
12    WHERE field_id = p_field_id;  
13  
14    RAISE NOTICE 'Temperature is too high. Crop status updated.';  
15   ELSIF p_moisture < 40 THEN  
16    UPDATE crops  
17    SET status = 'Low Moisture Risk'  
18    WHERE field_id = p_field_id;  
19  
20    RAISE NOTICE 'Moisture is too low. Crop status updated.';  
21   ELSE  
22    UPDATE crops  
23    SET status = 'Healthy'  
24    WHERE field_id = p_field_id;  
25  
26    RAISE NOTICE 'Field conditions are normal.';  
27   END IF;  
28 END;  
29 $$;  
30 CALL check_sensor_status(1, 35, 50);
```

The results pane shows the execution of the procedure with the following output:

```
1: CREATE  
Statement executed successfully
```

A notification message is displayed: "Temperature is too high. Crop status updated."

3, This procedure inserts a new account record into the users table using the provided username, password, full name, and role ID.



The screenshot shows a SQL IDE interface with a code editor, a toolbar, and a results pane. The code editor contains a PL/SQL procedure named `add_new_user` that takes `p_username`, `p_password`, `p_full_name`, and `p_role_id` as parameters. The procedure inserts a new record into the `users` table. The results pane shows the execution of the procedure with the following output:

```
1 CREATE OR REPLACE PROCEDURE add_new_user(  
2   p_username TEXT,  
3   p_password TEXT,  
4   p_full_name TEXT,  
5   p_role_id BIGINT  
6 )  
7 LANGUAGE plpgsql  
8 AS $$  
9 BEGIN  
10  
11   INSERT INTO users(  
12     username,  
13     password,  
14     full_name,  
15     role_id  
16   )  
17   VALUES (  
18     p_username,  
19     p_password,  
20     p_full_name,  
21     p_role_id  
22   );  
23  
24   RAISE NOTICE 'New user added successfully.';  
25 END;  
26 $$;  
27 CALL add_new_user('zehra', '1234', 'Zehra Tas', 1);
```

The results pane shows the execution of the procedure with the following output:

```
1: CREATE 2: CALL  
Statement executed successfully
```

A notification message is displayed: "New user added successfully."

4, This function returns the total number of crops in a given field for analytical purposes.

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```
1 CREATE OR REPLACE FUNCTION get_crop_count(p_field_id INT)
2 RETURNS INT
3 LANGUAGE plpgsql
4 AS $$
5 DECLARE total INT;
6 BEGIN
7     SELECT COUNT(*) INTO total
8     FROM crops
9     WHERE field_id = p_field_id;
10
11     RETURN total;
12 END;
13 $$;
14 SELECT get_crop_count(1);
```

 Connected (2 queries)

 Run | Explain Analyze 205ms 1 row

1: CREATE 2: SELECT 1

| # | get_crop_count |
|---|----------------|
| 1 | 0              |

5, TRIGGER Detects low moisture levels and warns the system to prevent crop damage.

Untitled Save Primary Active neondb  

```
1 CREATE OR REPLACE FUNCTION low_moisture_warning()
2 RETURNS TRIGGER AS $$
3 BEGIN
4     IF NEW.moisture < 60 THEN
5         RAISE NOTICE 'Warning: Low moisture detected!';
6     END IF;
7
8     RETURN NEW;
9 END;
10 $$ LANGUAGE plpgsql;
11 CREATE TRIGGER trg_low_moisture
12 AFTER INSERT ON sensorreadings
13 FOR EACH ROW
14 EXECUTE FUNCTION low_moisture_warning();
```

 Connected (2 queries)

 Run | Explain Analyze 201ms No result

1: CREATE 2: CREATE

Statement executed successfully

## SQL Editor

production

Saved History

insert new sensor reading  
May 16, 2026 - 4:51pm

insert sensor reading for field 2  
May 16, 2026 - 4:51pm

trigger for low moisture warning on sensor readings  
May 16, 2026 - 4:50pm

low moisture function  
May 16, 2026 - 4:50pm

Untitled Save

Primary Active neondb

```

1 INSERT INTO sensorreadings(field_id, temperature, moisture, reading_date)
2 VALUES (2, 25, 40, NOW());

```

Connected (1 query)

Run Explain Analyze

Warning: Low moisture detected!

6, This trigger prevents insertion of farms with duplicate names to maintain data integrity.

Untitled Save

Primary Active neondb

```

1 CREATE OR REPLACE FUNCTION prevent_duplicate_farm()
2 RETURNS TRIGGER
3 LANGUAGE plpgsql
4 AS $$
5 BEGIN
6     IF EXISTS (
7         SELECT 1 FROM farms WHERE farm_name = NEW.farm_name
8     ) THEN
9         RAISE EXCEPTION 'Farm name already exists!';
10    END IF;
11
12    RETURN NEW;
13 END;
14 $$;
15
16 CREATE TRIGGER trg_unique_farm
17 BEFORE INSERT ON farms
18 FOR EACH ROW
19 EXECUTE FUNCTION prevent_duplicate_farm();

```

Connected (2 queries)

Run Explain Analyze 190ms No result

1: CREATE 2: CREATE

Statement executed successfully



7, This trigger logs a "Critical temperature detected!" message into an alerts table whenever a new sensor reading exceeds 30 degrees.

Untitled

Save

Primary Active

neondb

```
1 CREATE TABLE alerts (  
2   alert_id BIGINT GENERATED ALWAYS AS IDENTITY PRIMARY KEY,  
3   field_id BIGINT,  
4   message TEXT,  
5   created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP  
6 );  
7 CREATE OR REPLACE FUNCTION high_temperature_alert()  
8 RETURNS TRIGGER AS $$  
9 BEGIN  
10  IF NEW.temperature > 30 THEN  
11    INSERT INTO alerts(field_id, message)  
12    VALUES (  
13      NEW.field_id,  
14      'Critical temperature detected!'  
15    );  
16  END IF;  
17  RETURN NEW;  
18 END;  
19 $$ LANGUAGE plpgsql;  
20 CREATE TRIGGER trg_high_temperature  
21 AFTER INSERT ON sensorreadings  
22 FOR EACH ROW  
23 EXECUTE FUNCTION high_temperature_alert();  
24
```

Connected (3 queries)

Run

Explain

Analyze

161ms No result

1: CREATE 2: CREATE 3: CREATE

Statement executed successfully

8, This PL/pgSQL function iterates through every field in the database, calculates their average temperature and moisture levels from recent sensor readings, and executes a status check procedure to evaluate and update their crop conditions.

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```
1 CREATE OR REPLACE FUNCTION monitor_all_fields()
2 RETURNS VOID AS $$
3 DECLARE
4     field_record RECORD;
5     avg_temp NUMERIC;
6     avg_moisture NUMERIC;
7 BEGIN
8     FOR field_record IN SELECT field_id FROM fields LOOP
9
10        SELECT AVG(temperature), AVG(moisture)
11        INTO avg_temp, avg_moisture
12        FROM SensorReadings
13        WHERE field_id = field_record.field_id;
14
15        IF avg_temp IS NOT NULL THEN
16            CALL check_sensor_status(field_record.field_id, avg_temp, avg_moisture);
17        END IF;
18    END LOOP;
19 END;
20 $$ LANGUAGE plpgsql;
21
22 SELECT monitor_all_fields();
```

Connected (2 queries)

▶ Run

Explain

Analyze

219ms1 row

1: CREATE2: SELECT 1

| # | monitor_all_fields |
|---|--------------------|
| 1 |                    |

⚠ Field conditions are normal.

⚠ Field conditions are normal.

⌂⬇🖨

## 4)DB Connection

This section explains how the application connects to the PostgreSQL database using **Npgsql**.

The connection is established using a connection string obtained from the cloud database provider.

A simple SQL query is executed to verify user login credentials (username and password).

```
24 private void btnLogin_Click(object sender, EventArgs e)
25 {
26     using (var conn = new NpgsqlConnection(connString))
27     {
28         conn.Open();
29
30         string query = @"SELECT * FROM Users
31             WHERE username=@u AND password=@p";
```

```
9 namespace SmartFarmingSystem
10 {
11     E Başvuru
12     public partial class LoginForm : Form
13     {
14         string connString = "Host=ep-bold-surf-apre6yz3.c-7.us-east-1.aws.neon.tech;Database=neondb;Username=neondb_owner;Password=npq_nqxUsDFfP10g;SSL Mode=Require;Trust Server Certificate=true;";
```

### Connect your app manually

Copy and paste your production branch's connection details into your app's .env file

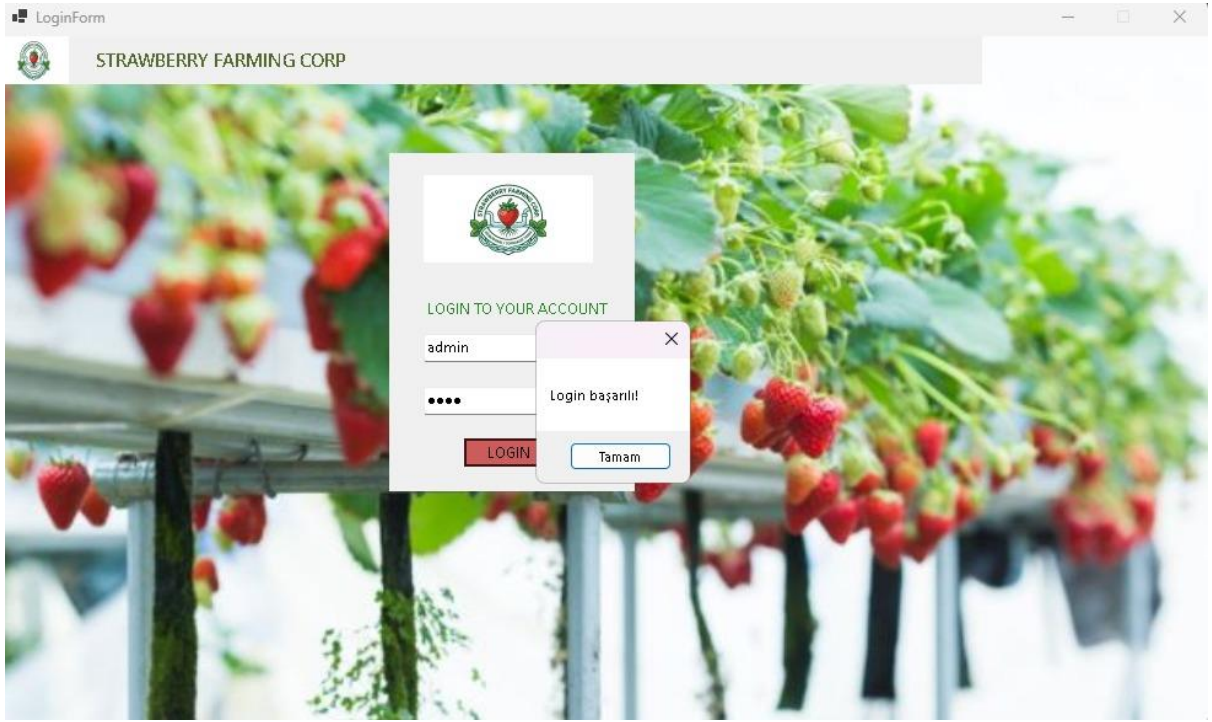
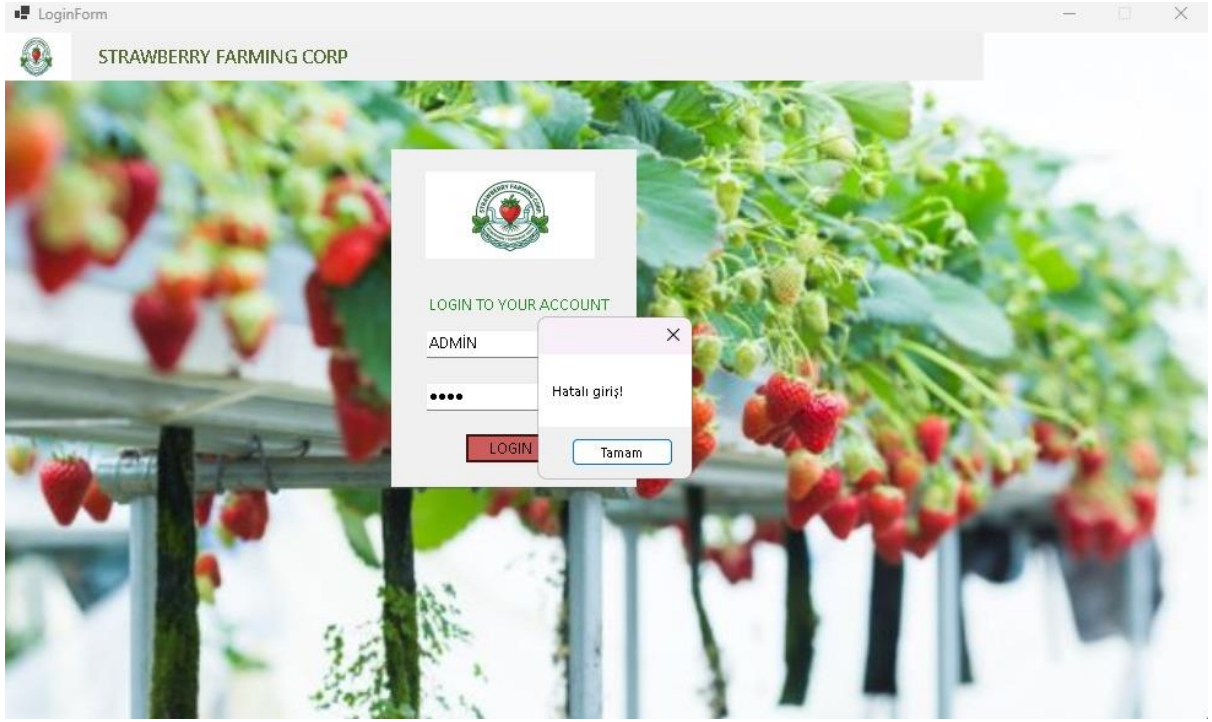
Connection string

Connection parameters

|             |                                                          |
|-------------|----------------------------------------------------------|
| Host        | ep-bold-surf-apre6yz3.c-7.us-east-1.aws.neon.tech        |
| Database    | neondb                                                   |
| Role        | neondb_owner                                             |
| Password    | *****                                                    |
| Pooler host | ep-bold-surf-apre6yz3-pooler.c-7.us-east-1.aws.neon.tech |

## **5)Codes and GUI**

Login Page



```
using System;
using System.Windows.Forms;
using Npgsql;

namespace SmartFarmingSystem
{
    public partial class LoginForm : Form
    {
        string connectionString = "Host=ep-bold-surf-apretyr3-c-7-us-east-1.amazonaws.com;Database=neondb;Username=neondb_owner;Password=ngp_nqUs0FFp1Bg;SSL Mode=Require;Trust Server Certificate=true;";

        public LoginForm()
        {
            InitializeComponent();
            this.StartPosition = FormStartPosition.CenterScreen;
            this.Size = new Size(1000, 600);
            this.FormBorderStyle = FormBorderStyle.FixedSingle;
            this.MaximizeBox = false;
        }

        private void btnLogin_Click(object sender, EventArgs e)
        {
            using (var conn = new NpgsqlConnection(connectionString))
            {
                conn.Open();

                string query = @"SELECT * FROM Users
                                WHERE username=@u AND password=@p";

                using (var cmd = new NpgsqlCommand(query, conn))
                {
                    cmd.Parameters.AddWithValue("@u", txtUsername.Text);
                    cmd.Parameters.AddWithValue("@p", txtPassword.Text);

                    var reader = cmd.ExecuteReader();

                    if (reader.Read())
                    {
                        MessageBox.Show("Login başarılı!");

                        Dashboard d = new Dashboard();
                        d.Show();
                        this.Hide();
                    }
                    else
                    {
                        MessageBox.Show("Hatalı giriş!");
                    }
                }
            }
        }
    }
}
```

## Dashboard

 STRAWBERRY FARMING CORP

Welcome! Have a nice day

WHAT YOU WANT TO DO?

Manage Crops

Manage Farms

Manage Sensors

Manage Field

Statistics

Log Out

Crops: 3

Farms: 2

|   | field_name       | temperature | moisture | reading_date     |
|---|------------------|-------------|----------|------------------|
| ▶ | Greenhouse A     | 22          | 70       | 6.05.2026 10:00  |
|   | Greenhouse B     | 22          | 70       | 6.05.2026 10:00  |
|   | Vertical Tower 1 | 23          | 54       | 14.05.2026 16:09 |
| * |                  |             |          |                  |

```
SmartFarmingSystem SmartFarmingSystem.Dashboard connString Dashboard.cs
1 using System;
2 using System.Collections.Generic;
3 using System.ComponentModel;
4 using System.Data;
5 using System.Drawing;
6 using System.Text;
7 using System.Windows.Forms;
8 using Npgsql;
9
10 namespace SmartFarmingSystem
11 {
12     using System.Data;
13
14     public partial class Dashboard : Form
15     {
16         string connString = "Host=ep-bold-surf-apr6y23.c-7.us-east-1.aws.neon.tech;Database=neondb;Username=neondb_owner;Password=npg_nqxUsDffP18g;SSL Mode=Require;Trust Server Certificate=true;";
17
18         6 bagpuru
19         public Dashboard()
20         {
21             InitializeComponent();
22             this.StartPosition = FormStartPosition.CenterScreen;
23             this.Size = new Size(1000, 600);
24             this.FormBorderStyle = FormBorderStyle.FixedSingle;
25             this.MaximizeBox = false;
26         }
27
28         1 bagpuru
29         private void Dashboard_Load(object sender, EventArgs e)
30         {
31             LoadDashboardData();
32         }
33
34         1 bagpuru
35         void LoadDashboardData()
36         {
37             using (var conn = new NpgsqlConnection(connString))
38             {
39                 conn.Open();
40
41                 // toplan crops
42                 using (var cmd = new NpgsqlCommand("SELECT COUNT(*) FROM Crops", conn))
43                 {
44                     lblCrops.Text = cmd.ExecuteScalar().ToString();
45                 }
46
47                 // toplan farms
48                 using (var cmd = new NpgsqlCommand("SELECT COUNT(*) FROM Farms", conn))
49                 {
50                     lblFarms.Text = cmd.ExecuteScalar().ToString();
51                 }
52             }
53         }
54
55         46
56         using (var cmd = new NpgsqlCommand("SELECT COUNT(*) FROM Farms", conn))
57         {
58             lblFarms.Text = cmd.ExecuteScalar().ToString();
59         }
60
61         // sensör tablosu
62         string query = @"SELECT f.field_name, s.temperature, s.moisture, s.reading_date
63             FROM SensorReadings s
64             JOIN Fields f ON s.field_id = f.field_id";
65
66         56
67         using (var da = new NpgsqlDataAdapter(query, conn))
68         {
69             DataTable dt = new DataTable();
70             da.Fill(dt);
71             dataGridViewtbl1.DataSource = dt;
72         }
73
74         1 bagpuru
75         private void lbl3_Click(object sender, EventArgs e)
76         {
77         }
78
79         1 bagpuru
80         private void panel2_Paint(object sender, PaintEventArgs e)
81         {
82         }
83
84         1 bagpuru
85         private void button2_Click(object sender, EventArgs e)
86         {
87             FarmsForm f = new FarmsForm(); // şimdilik bunu aç
88             f.Show();
89             this.Close();
90         }
91
92         1 bagpuru
93         private void button4_Click(object sender, EventArgs e)
94         {
95             FieldsForm f = new FieldsForm();
96             f.Show();
97             this.Close();
98         }
99     }
100 }
```

Crops management



Adding crop

CropsForm

STRAWBERRY FARMING CORP

Welcome! Have a nice day

CROPS MANAGEMENT

Crop Name: Strawberry test

Field: Greenhouse A

Date: 14 Mayıs 2026 Perşembe

ADD

UPDATE

DELETE

BACK

|   | crop_id | crop_name           | field_name       | planting_date | status    |
|---|---------|---------------------|------------------|---------------|-----------|
| ▶ | 2       | Strawberry Festi... | Greenhouse B     | 15.02.2026    | Ready     |
|   | 3       | Strawberry San ...  | Vertical Tower 1 | 10.03.2026    | Planted   |
|   | 1       | Strawberry Albi...  | Greenhouse A     | 1.03.2026     | Harvested |
|   | 18      | Strawberry test     | Greenhouse A     | 14.05.2026    | New       |
| * |         |                     |                  |               |           |

Updating crop

CropsForm

STRAWBERRY FARMING CORP

Welcome! Have a nice day

CROPS MANAGEMENT

Crop Name: Strawberry test

Field: Greenhouse B

Date: 14 Mayıs 2026 Perşembe

ADD

UPDATE

DELETE

BACK

|   | crop_id | crop_name           | field_name       | planting_date | status    |
|---|---------|---------------------|------------------|---------------|-----------|
| ▶ | 2       | Strawberry Festi... | Greenhouse B     | 15.02.2026    | Ready     |
|   | 3       | Strawberry San ...  | Vertical Tower 1 | 10.03.2026    | Planted   |
|   | 1       | Strawberry Albi...  | Greenhouse A     | 1.03.2026     | Harvested |
|   | 18      | Strawberry test     | Greenhouse B     | 14.05.2026    | Updated   |
| * |         |                     |                  |               |           |

Deleting crop



CropsForm

 STRAWBERRY FARMING CORP

Welcome! Have a nice day

 CROPS MANAGEMENT

Crop Name:

Field:

Date:

|   | crop_id | crop_name           | field_name       | planting_date | status    |
|---|---------|---------------------|------------------|---------------|-----------|
| ▶ | 2       | Strawberry Festi... | Greenhouse B     | 15.02.2026    | Ready     |
|   | 3       | Strawberry San ...  | Vertical Tower 1 | 10.03.2026    | Planted   |
|   | 1       | Strawberry Albi...  | Greenhouse A     | 1.03.2026     | Harvested |
| • |         |                     |                  |               |           |

```
CropsForm.cs | FieldsForm.cs
SmartFarmingSystem | SmartFarmingSystem.CropsForm | connString

1  using System;
2  using System.Collections.Generic;
3  using System.ComponentModel;
4  using System.Data;
5  using System.Drawing;
6  using System.Linq;
7  using System.Windows.Forms;
8  using Npgsql;
9
10 namespace SmartFarmingSystem
11 {
12     public partial class CropsForm : Form
13     {
14         string connString = "Host=ep-bold-surf-apre6yz3.c-7.us-east-1.aws.neon.tech;Database=neondb;Username=neondb_owner;Password=npg_nqxUsDfP10g;SSL Mode=Require;Trust Server Certificate=true;";
15         int selectedCropId = -1;
16
17         public CropsForm()
18         {
19             InitializeComponent();
20             this.Load += CropsForm_Load;
21             this.StartPosition = FormStartPosition.CenterScreen;
22             this.Size = new Size(1000, 600);
23             this.FormBorderStyle = FormBorderStyle.FixedSingle;
24             this.MaximizeBox = false;
25         }
26
27         void LoadCrops()
28         {
29             using (var conn = new NpgsqlConnection(connString))
30             {
31                 conn.Open();
32
33                 string query = @"SELECT c.crop_id, c.crop_name, f.field_name, c.planting_date, c.status
34                                FROM crops c
35                                JOIN fields f ON c.field_id = f.field_id";
36
37                 using (var da = new NpgsqlDataAdapter(query, conn))
38                 {
39                     DataTable dt = new DataTable();
40                     da.Fill(dt);
41                     dataGridView1.DataSource = dt;
42                 }
43             }
44         }
45
46         void LoadFields()
47         {
48             cmbField.Items.Clear();
49             cmbField.Items.Add("Greenhouse A");
50             cmbField.Items.Add("Greenhouse B");
51             cmbField.Items.Add("Vertical Tower 1");
52         }
53     }
54 }
```

```

46         cmbField.Items.Add("Greenhouse A");
47         cmbField.Items.Add("Greenhouse B");
48         cmbField.Items.Add("Vertical Tower 1");
49     }
50 }
51
52 private void lbl6_Click(object sender, EventArgs e)
53 {
54 }
55
56
57
58 private void btnAdd_Click(object sender, EventArgs e)
59 {
60     if (txtCropName1.Text == "" || cmbField.Text == "")
61     {
62         MessageBox.Show("Boş alan bırakma!");
63         return;
64     }
65     using (var conn = new NpgsqlConnection(connString))
66     {
67         conn.Open();
68
69         string query = @"INSERT INTO crops (field_id, crop_name, planting_date, status)
70 VALUES (
71     (SELECT field_id FROM fields WHERE field_name=@f LIMIT 1),
72     @name, @date, @status
73 );";
74
75         using (var cmd = new NpgsqlCommand(query, conn))
76         {
77             cmd.Parameters.AddWithValue("@f", cmbField.Text);
78             cmd.Parameters.AddWithValue("@name", txtCropName1.Text);
79             cmd.Parameters.AddWithValue("@date", dtPlantingDate.Value);
80             cmd.Parameters.AddWithValue("@status", "New");
81
82             cmd.ExecuteNonQuery();
83         }
84     }
85     LoadCrops();
86 }
87
88
89
90 private void CropsForm_Load(object sender, EventArgs e)
91 {
92     LoadCrops();
93     LoadFields();
94 }
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```

```

CropsForm.cs x FieldsForm.cs
SmartFarmingSystem SmartFarmingSystem.CropsForm connString
97
98 private void dataGridView1_CellClick(object sender, DataGridViewCellEventArgs e)
99 {
100     if (e.RowIndex >= 0)
101     {
102         var row = dataGridView1.Rows[e.RowIndex];
103
104         // ID
105         if (row.Cells[0].Value != DBNull.Value)
106             selectedCropId = Convert.ToInt32(row.Cells[0].Value);
107
108         // Text alanları
109         txtCropName1.Text = row.Cells[1].Value?.ToString();
110         cmbField.Text = row.Cells[2].Value?.ToString();
111
112         // Date fix (EN KÜTÜK WİSTİM)
113         var v = row.Cells[3].Value;
114
115         if (v != DBNull.Value)
116         {
117             if (v is DateTime dt)
118                 dtPlantingDate.Value = dt;
119
120             else if (v is DateOnly d)
121                 dtPlantingDate.Value = d.ToDateTime(TimeOnly.MinValue);
122         }
123     }
124 }
125
126
127 private void btnUpdate_Click(object sender, EventArgs e)
128 {
129     if (selectedCropId == -1)
130     {
131         MessageBox.Show("Önce listeden bir kayıt seç!");
132         return;
133     }
134
135     using (var conn = new NpgsqlConnection(connString))
136     {
137         conn.Open();
138
139         string query = @"UPDATE crops
140 SET crop_name=@name,
141     planting_date=@date,
142     field_id = (SELECT field_id FROM fields WHERE field_name=@f LIMIT 1),
143     status=@status
144 WHERE crop_id=@id";
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```

```

144         where crop_id=@id";
145
146         using (var cmd = new NpgsqlCommand(query, conn))
147         {
148             cmd.Parameters.AddWithValue("@name", txtCropName1.Text);
149             cmd.Parameters.AddWithValue("@date", dtPlantingDate.Value);
150             cmd.Parameters.AddWithValue("@f", cmbField.Text);
151             cmd.Parameters.AddWithValue("@status", "updated");
152             cmd.Parameters.AddWithValue("@id", selectedCropId);
153
154             int affected = cmd.ExecuteNonQuery();
155
156             if (affected == 0)
157                 MessageBox.Show("Güncellenecek kayıt bulunamadı!");
158         }
159
160         selectedCropId = -1; // reset
161         LoadCrops();
162     }
163
164
165     1 bayguru
166     private void btnDelete_Click(object sender, EventArgs e)
167     {
168         if (dataGridView1.CurrentRow != null)
169         {
170             int id = Convert.ToInt32(dataGridView1.CurrentRow.Cells[0].Value);
171
172             using (var conn = new NpgsqlConnection(connString))
173             {
174                 conn.Open();
175
176                 string query = "DELETE FROM crops WHERE crop_id=@id";
177
178                 using (var cmd = new NpgsqlCommand(query, conn))
179                 {
180                     cmd.Parameters.AddWithValue("@id", id);
181                     cmd.ExecuteNonQuery();
182                 }
183             }
184
185             LoadCrops();
186         }
187     }
188
189     1 bayguru
190     private void dataGridView1_CellContentClick(object sender, DataGridViewCellEventArgs e)
191     {
192     }
193
194     1 bayguru
195     private void btnBack_Click(object sender, EventArgs e)
196     {
197         Dashboard d = new Dashboard();
198         d.Show();
199         this.Close();
200     }

```

## Farms management

### Adding farm



STRAWBERRY FARMING CORP

Welcome! Have a nice day



FARM MANAGEMENT

Farm:

Location:

Size:

ADD

UPDATE

DELETE

BACK

|   | farm_id | farm_name         | location | size_acre |
|---|---------|-------------------|----------|-----------|
| ▶ | 1       | Hydro Strawber... | Nicosia  | 12        |
|   | 2       | Smart Berry Lab   | Kyrenia  | 10        |
|   | 10      | Hydro test        | ciu      | 15        |
| * |         |                   |          |           |

### Updating farm

FarmsForm

STRAWBERRY FARMING CORP

Welcome! Have a nice day



FARM MANAGEMENT

Farm:

Location:

Size:

ADD

UPDATE

DELETE

BACK

|   | farm_id | farm_name         | location | size_acre |
|---|---------|-------------------|----------|-----------|
| ▶ | 1       | Hydro Strawber... | Nicosia  | 12        |
|   | 2       | Smart Berry Lab   | Kýrenia  | 10        |
|   | 10      | Hydro test2       | ciu st   | 19        |
| * |         |                   |          |           |

Deleting farm

FarmsForm

STRAWBERRY FARMING CORP

Welcome! Have a nice day



FARM MANAGEMENT

Farm:

Location:

Size:

ADD

UPDATE

DELETE

BACK

|   | farm_id | farm_name         | location | size_acre |
|---|---------|-------------------|----------|-----------|
| ▶ | 1       | Hydro Strawber... | Nicosia  | 12        |
|   | 2       | Smart Berry Lab   | Kýrenia  | 10        |
| * |         |                   |          |           |

```
FarmsForm.cs X
SmartFarmingSystem SmartFarmingSystem.FarmsForm connString
1 using Npgsql;
2 using System;
3 using System.Collections.Generic;
4 using System.ComponentModel;
5 using System.Data;
6 using System.Drawing;
7 using System.Text;
8 using System.Windows.Forms;
9 namespace SmartFarmingSystem
10 {
11     5 baguruu
12     public partial class FarmsForm : Form
13     {
14         string connString = "Host=ep-bold-surf-apr6yz3.c-7.us-east-1.amazonaws.com;Database=neondb;Username=neondb_owner;Password=npq_nqxusDFFP10g;SSL Mode=Require;Trust Server Certificate=true;";
15         public FarmsForm()
16         {
17             InitializeComponent();
18             this.Load += FarmsForm_Load;
19             btnAdd.Click += btnAdd_Click;
20             btnUpdate.Click += btnUpdate_Click;
21             btnDelete.Click += btnDelete_Click;
22             btnBack.Click += btnBack_Click;
23             dataGridView1.CellClick += dataGridView1_CellClick;
24         }
25     }
26
27     // LOAD
28     4 baguruu
29     void LoadFarms()
30     {
31         using (var conn = new NpgsqlConnection(connString))
32         {
33             conn.Open();
34
35             string query = "SELECT * FROM farms";
36
37             using (var da = new NpgsqlDataAdapter(query, conn))
38             {
39                 DataTable dt = new DataTable();
40                 da.Fill(dt);
41                 dataGridView1.DataSource = dt;
42             }
43         }
44     }
45
46     1 baguruu
47     private void FarmsForm_Load(object sender, EventArgs e)
48     {
49         LoadFarms();
50     }
51 }
```

```
FarmsForm.cs X
SmartFarmingSystem SmartFarmingSystem.FarmsForm connString
51 // ADD
52 private void btnAdd_Click(object sender, EventArgs e)
53 {
54     if (txtFarmName.Text == "" || txtLocation.Text == "" || txtSize.Text == "")
55     {
56         MessageBox.Show("Boş alan bırakma!");
57         return;
58     }
59
60     using (var conn = new NpgsqlConnection(connString))
61     {
62         conn.Open();
63
64         string query = "INSERT INTO farms (farm_name, location, size_acre) VALUES (@n, @l, @s)";
65
66         using (var cmd = new NpgsqlCommand(query, conn))
67         {
68             cmd.Parameters.AddWithValue("@n", txtFarmName.Text);
69             cmd.Parameters.AddWithValue("@l", txtLocation.Text);
70             cmd.Parameters.AddWithValue("@s", int.Parse(txtSize.Text));
71
72             try
73             {
74                 cmd.ExecuteNonQuery();
75             }
76             catch
77             {
78                 MessageBox.Show("Bu farm zaten var!");
79             }
80         }
81
82         LoadFarms();
83     }
84
85     // UPDATE
86     private void btnUpdate_Click(object sender, EventArgs e)
87     {
88         if (dataGridView1.CurrentRow != null)
89         {
90             int id = Convert.ToInt32(dataGridView1.CurrentRow.Cells[0].Value);
91
92             using (var conn = new NpgsqlConnection(connString))
93             {
94                 conn.Open();
95
96                 string query = @"UPDATE farms
97                     SET farm_name=@n, location=@l, size_acre=@s";
98             }
99         }
100     }
101 }
```

```
FarmsForm.cs
SmartFarmingSystem
SmartFarmingSystem.FarmsForm
connString

84
85 // UPDATE
86 1 backup
87 private void btnUpdate_Click(object sender, EventArgs e)
88 {
89     if (dataGridView1.CurrentRow != null)
90     {
91         int id = Convert.ToInt32(dataGridView1.CurrentRow.Cells[0].Value);
92         using (var conn = new NpgsqlConnection(connString))
93         {
94             conn.Open();
95
96             string query = @"UPDATE farms
97                             SET farm_name=@n, location=@l, size_acre=@s
98                             WHERE farm_id=@id";
99
100             using (var cmd = new NpgsqlCommand(query, conn))
101             {
102                 cmd.Parameters.AddWithValue("@n", txtFarmName.Text);
103                 cmd.Parameters.AddWithValue("@l", txtLocation.Text);
104                 cmd.Parameters.AddWithValue("@s", int.Parse(txtSize.Text));
105                 cmd.Parameters.AddWithValue("@id", id);
106
107                 cmd.ExecuteNonQuery();
108             }
109         }
110         LoadFarms();
111     }
112 }
113
114 // DELETE
115 1 backup
116 private void btnDelete_Click(object sender, EventArgs e)
117 {
118     if (dataGridView1.CurrentRow != null)
119     {
120         int id = Convert.ToInt32(dataGridView1.CurrentRow.Cells[0].Value);
121         using (var conn = new NpgsqlConnection(connString))
122         {
123             conn.Open();
124
125             string query = "DELETE FROM farms WHERE farm_id=@id";
126
127             using (var cmd = new NpgsqlCommand(query, conn))
128             {
129                 cmd.Parameters.AddWithValue("@id", id);
130                 cmd.ExecuteNonQuery();
131             }
132         }
133     }
134 }
```

```
FarmsForm.cs
SmartFarmingSystem
SmartFarmingSystem.FarmsForm
connString

131         cmd.ExecuteNonQuery();
132     }
133 }
134 LoadFarms();
135 }
136 }
137 }
138
139 // GRID CLICK
140 private void dataGridView1_CellClick(object sender, DataGridViewCellEventArgs e)
141 {
142     if (e.RowIndex >= 0)
143     {
144         var row = dataGridView1.Rows[e.RowIndex];
145
146         txtFarmName.Text = row.Cells[1].Value.ToString();
147         txtLocation.Text = row.Cells[2].Value.ToString();
148         txtSize.Text = row.Cells[3].Value.ToString();
149     }
150 }
151
152 // BACK
153 private void btnBack_Click(object sender, EventArgs e)
154 {
155     this.Hide();
156     Dashboard d = new Dashboard();
157     d.Show();
158     this.Close();
159 }
160
161 private void label7_Click(object sender, EventArgs e)
162 {
163 }
164
165 private void label2_Click(object sender, EventArgs e)
166 {
167 }
168
169 }
170
171
172
173
174 }
```



Sensor management

Adding sensor

SensorReadingsForm

STRAWBERRY FARMING CORP

Welcome! Have a nice day

 SENSOR MANAGEMENT

Temperature:

Moisture:

Field: 

▼

Date: 14 Mayıs 2026 Perşembe 

📅

ADD

UPDATE

DELETE

BACK

|   | reading_id | field_name       | temperature | moisture | reading_date     |
|---|------------|------------------|-------------|----------|------------------|
| ▶ | 1          | Greenhouse A     | 22          | 70       | 6.05.2026 10:00  |
|   | 2          | Greenhouse B     | 22          | 70       | 6.05.2026 10:00  |
|   | 3          | Vertical Tower 1 | 23          | 54       | 14.05.2026 16:09 |
|   | 16         | Greenhouse B     | 24          | 80       | 14.05.2026 16:44 |
| ✱ |            |                  |             |          |                  |

Updating sensor

SensorReadingsForm

STRAWBERRY FARMING CORP

Welcome! Have a nice day

 SENSOR MANAGEMENT

Temperature: 24

Moisture: 80

Field: 

Greenhouse B ▼

Date: 14 Mayıs 2026 Perşembe 

📅

ADD

UPDATE

DELETE

BACK

|   | reading_id | field_name       | temperature | moisture | reading_date     |
|---|------------|------------------|-------------|----------|------------------|
| ▶ | 1          | test field       | 22          | 70       | 6.05.2026 10:00  |
|   | 2          | Greenhouse B     | 22          | 70       | 6.05.2026 10:00  |
|   | 3          | Vertical Tower 1 | 23          | 54       | 14.05.2026 16:09 |
|   | 16         | Greenhouse B     | 24          | 80       | 14.05.2026 16:44 |
| ✱ |            |                  |             |          |                  |



## Deleting sensor

SensorReadingsForm

STRAWBERRY FARMING CORP

Welcome! Have a nice day

SENSOR MANAGEMENT

Temperature: 24

Moisture: 80

Field: Greenhouse B

Date: 14 Mayıs 2026 Perşembe

ADD UPDATE DELETE

BACK

|   | reading_id | field_name       | temperature | moisture | reading_date     |
|---|------------|------------------|-------------|----------|------------------|
| ▶ | 1          | test field       | 22          | 70       | 6.05.2026 10:00  |
|   | 2          | Greenhouse B     | 22          | 70       | 6.05.2026 10:00  |
|   | 3          | Vertical Tower 1 | 23          | 54       | 14.05.2026 16:09 |
| * |            |                  |             |          |                  |

```
1 using System;
2 using System.Collections.Generic;
3 using System.ComponentModel;
4 using System.Data;
5 using System.Drawing;
6 using System.Linq;
7 using System.Windows.Forms;
8 using Npgsql;
9
10 namespace SmartFarmingSystem
11 {
12     3 bayyuru
13     public partial class SensorReadingsForm : Form
14     {
15         int selectedReadingId = 0;
16         string connString = "Host=ep-bold-surf-apre6yz3.c-7.us-east-1.aws.neon.tech;Database=neondb;Username=neondb_owner;Password=npq_nqxusDFfP1Bg;SSL Mode=Require;Trust Server Certificate=true;";
17
18         4 bayyuru
19         void LoadSensors()
20         {
21             using (var conn = new NpgsqlConnection(connString))
22             {
23                 conn.Open();
24
25                 string query = @"SELECT s.reading_id, f.field_name, s.temperature, s.moisture, s.reading_date
26                     FROM sensorreadings s
27                     JOIN fields f ON s.field_id = f.field_id";
28
29                 using (var da = new NpgsqlDataAdapter(query, conn))
30                 {
31                     DataTable dt = new DataTable();
32                     da.Fill(dt);
33                     dataGridView1.DataSource = dt;
34                 }
35             }
36
37             1 bayyuru
38             void LoadFields()
39             {
40                 using (var conn = new NpgsqlConnection(connString))
41                 {
42                     conn.Open();
43
44                     string q = "SELECT field_id, field_name FROM fields";
45
46                     using (var da = new NpgsqlDataAdapter(q, conn))
47                     {
48                         DataTable dt = new DataTable();
49                         da.Fill(dt);
50                         cmbField.DataSource = dt;
51                     }
52                 }
53             }
54         }
55     }
56 }
```

```
SensorReadingsForm.cs
SmartFarmingSystem
SmartFarmingSystem.SensorReadingsForm
LoadFields()

47
48 cmbField.DataSource = dt;
49 cmbField.DisplayMember = "field_name"; // görünen
50 cmbField.ValueMember = "field_id"; // arka plan (ID)
51
52
53
54
55 private void SensorReadingsForm_Load(object sender, EventArgs e)
56 {
57     LoadSensors();
58     LoadFields();
59 }
60
61 public SensorReadingsForm()
62 {
63     InitializeComponent();
64     this.Load += SensorReadingsForm_Load;
65     this.StartPosition = FormStartPosition.CenterScreen;
66     this.Size = new Size(1000, 600);
67     this.FormBorderStyle = FormBorderStyle.FixedSingle;
68     this.MaximizeBox = false;
69 }
70
71 private void btnAdd_Click(object sender, EventArgs e)
72 {
73     using (var conn = new NpgsqlConnection(connString))
74     {
75         conn.Open();
76
77         string query = @"INSERT INTO sensorreadings (field_id, temperature, moisture, reading_date)
78 VALUES (
79     (SELECT field_id FROM fields WHERE field_name=@f),
80     @temp, @moist, @date
81 )";
82
83         using (var cmd = new NpgsqlCommand(query, conn))
84         {
85             cmd.Parameters.AddWithValue("@f", cmbField.Text);
86             cmd.Parameters.AddWithValue("@temp", int.Parse(txtTemperature.Text));
87             cmd.Parameters.AddWithValue("@moist", int.Parse(txtMoisture.Text));
88             cmd.Parameters.AddWithValue("@date", dtReadingDate.Value);
89
90             cmd.ExecuteNonQuery();
91         }
92     }
93
94     LoadSensors();
95     txtTemperature.Clear();
96 }
```

```
SensorReadingsForm.cs
SmartFarmingSystem
SmartFarmingSystem.SensorReadingsForm
LoadFields()

93 LoadSensors();
94 txtTemperature.Clear();
95 txtMoisture.Clear();
96 cmbField.SelectedIndex = -1;
97
98
99 private void btnDelete_Click(object sender, EventArgs e)
100 {
101
102     if (dataGridView1.CurrentRow != null)
103     {
104         int id = Convert.ToInt32(dataGridView1.CurrentRow.Cells["reading_id"].Value);
105
106         using (var conn = new NpgsqlConnection(connString))
107         {
108             conn.Open();
109
110             string query = "DELETE FROM sensorreadings WHERE reading_id=@id";
111
112             using (var cmd = new NpgsqlCommand(query, conn))
113             {
114                 cmd.Parameters.AddWithValue("@id", id);
115                 cmd.ExecuteNonQuery();
116             }
117         }
118
119         LoadSensors();
120     }
121 }
122
123 private void btnUpdate_Click(object sender, EventArgs e)
124 {
125     if (selectedReadingId == 0)
126     {
127         MessageBox.Show("Lütfen bir kayıt seç!");
128         return;
129     }
130
131     if (cmbField.SelectedValue == null)
132     {
133         MessageBox.Show("Field seçili değil!");
134         return;
135     }
136
137     using (var conn = new NpgsqlConnection(connString))
138     {
139         conn.Open();
140
141         string q = @"UPDATE sensorreadings
142 SET field_id = @field_id, temperature = @temp, moisture = @moist, reading_date = @date
143 WHERE reading_id = @id";
144
145         using (var cmd = new NpgsqlCommand(q, conn))
146         {
147             cmd.Parameters.AddWithValue("@field_id", cmbField.SelectedValue);
148             cmd.Parameters.AddWithValue("@temp", int.Parse(txtTemperature.Text));
149             cmd.Parameters.AddWithValue("@moist", int.Parse(txtMoisture.Text));
150             cmd.Parameters.AddWithValue("@date", dtReadingDate.Value);
151             cmd.Parameters.AddWithValue("@id", selectedReadingId);
152             cmd.ExecuteNonQuery();
153         }
154     }
155
156     LoadSensors();
157 }
```

FieldsForm

STRAWBERRY FARMING CORP

Welcome! Have a nice day

 FIELDS MANAGEMENT

Farm: Smart Berry Lab

Field Name: test field2

Soil: Hydroponic2

Area: 55

ADD

UPDATE

DELETE

BACK

|   | field_id | farm_name         | field_name       | soil_type   | area |
|---|----------|-------------------|------------------|-------------|------|
| ▶ | 1        | Hydro Strawber... | Greenhouse A     | Hydroponic  | 3    |
|   | 2        | Hydro Strawber... | Greenhouse B     | Hydroponic  | 4    |
|   | 3        | Smart Berry Lab   | Vertical Tower 1 | Hydroponic  | 2    |
|   | 11       | Smart Berry Lab   | test field2      | Hydroponic2 | 55   |
| • |          |                   |                  |             |      |

## Updating Field

FieldsForm

 STRAWBERRY FARMING CORP

Welcome! Have a nice day

 FIELDS MANAGEMENT

Farm: Smart Berry Lab

Field Name: test field 11

Soil: Hydroponic22

Area: 5

ADDUPDATEDELETE

BACK

|   | field_id | farm_name         | field_name       | soil_type    | area |
|---|----------|-------------------|------------------|--------------|------|
| ▶ | 2        | Hydro Strawber... | Greenhouse B     | Hydroponic   | 4    |
|   | 3        | Smart Berry Lab   | Vertical Tower 1 | Hydroponic   | 2    |
|   | 1        | Smart Berry Lab   | Greenhouse A     | Hydroponic   | 3    |
|   | 11       | Smart Berry Lab   | test field 11    | Hydroponic22 | 5    |
| • |          |                   |                  |              |      |

## Deleting Field

FieldsForm

 STRAWBERRY FARMING CORP

Welcome! Have a nice day

 FIELDS MANAGEMENT

Farm: Smart Berry Lab

Field Name: test field 11

Soil: Hydroponic22

Area: 5

ADDUPDATEDELETE

BACK

|   | field_id | farm_name         | field_name       | soil_type  | area |
|---|----------|-------------------|------------------|------------|------|
| ▶ | 2        | Hydro Strawber... | Greenhouse B     | Hydroponic | 4    |
|   | 3        | Smart Berry Lab   | Vertical Tower 1 | Hydroponic | 2    |
|   | 1        | Smart Berry Lab   | Greenhouse A     | Hydroponic | 3    |
| • |          |                   |                  |            |      |

```
FieldForm.cs
SmartFarmingSystem
using Npgsql;
using System;
using System.Collections.Generic;
using System.ComponentModel;
using System.Data;
using System.Drawing;
using System.Text;
using System.Windows.Forms;

namespace SmartFarmingSystem
{
    public partial class FieldsForm : Form
    {
        string connString = "Host=ep-bold-surf-apre5yz3.c-7.us-east-1.aws.neon.tech;Database=neondb;Username=neondb_owner;Password=npg_usDFfp10g;SSL Mode=Require;Trust Server Certificate=true;";

        public FieldsForm()
        {
            InitializeComponent();
            this.Load += FieldsForm_Load;
            this.StartPosition = FormStartPosition.CenterScreen;

            this.FormBorderStyle = FormBorderStyle.FixedSingle;
            this.MaximizeBox = false;
            btnAdd.Click += btnAdd_Click;
            btnUpdate.Click += btnUpdate_Click;
            btnDelete.Click += btnDelete_Click;
            btnBack.Click += btnBack_Click;
            dataGridView1.CellClick += dataGridView1_CellClick;
        }

        // LOAD FARMS → dropdown
        void LoadFarms()
        {
            using (var conn = new NpgsqlConnection(connString))
            {
                conn.Open();

                string query = "SELECT farm_id, farm_name FROM farms";

                using (var da = new NpgsqlDataAdapter(query, conn))
                {
                    DataTable dt = new DataTable();
                    da.Fill(dt);

                    cmbFarm.DataSource = dt;
                    cmbFarm.DisplayMember = "farm_name";
                    cmbFarm.ValueMember = "farm_id";
                }
            }
        }
    }
}
```

```
FieldForm.cs
SmartFarmingSystem
// LOAD FIELDS
4 bapuru
void LoadFields()
{
    using (var conn = new NpgsqlConnection(connString))
    {
        conn.Open();

        string query = @"SELECT f.field_id, fa.farm_name, f.field_name, f.soil_type, f.area
                        FROM fields f
                        JOIN farms fa ON f.farm_id = fa.farm_id";

        using (var da = new NpgsqlDataAdapter(query, conn))
        {
            DataTable dt = new DataTable();
            da.Fill(dt);
            dataGridView1.DataSource = dt;
        }
    }
}

1 bapuru
private void FieldsForm_Load(object sender, EventArgs e)
{
    LoadFarms();
    LoadFields();
}

// + ADD
1 bapuru
private void btnAdd_Click(object sender, EventArgs e)
{
    if (txtFieldName.Text == "")
    {
        MessageBox.Show("Boş alan bırakma!");
        return;
    }

    using (var conn = new NpgsqlConnection(connString))
    {
        conn.Open();

        string query = @"INSERT INTO fields (farm_id, field_name, soil_type, area)
                        VALUES (@farm, @name, @soil, @area)";

        using (var cmd = new NpgsqlCommand(query, conn))
        {
            cmd.Parameters.AddWithValue("@farm", cmbFarm.SelectedValue);
            cmd.Parameters.AddWithValue("@name", txtFieldName.Text);
            cmd.Parameters.AddWithValue("@soil", txtSoil.Text);
        }
    }
}
```

```

FieldForm.cs
SmartFarmingSystem
SmartFarmingSystem.FieldForm
label2_Click(object sender, EventArgs e)

97      cmd.Parameters.AddWithValue("@name", txtFieldName.Text);
98      cmd.Parameters.AddWithValue("@soil", txtSoil.Text);
99      cmd.Parameters.AddWithValue("@area", int.Parse(txtArea.Text));
100
101      cmd.ExecuteNonQuery();
102  }
103  }
104
105      LoadFields();
106  }
107
108  // UPDATE
109  private void btnUpdate_Click(object sender, EventArgs e)
110  {
111      if (dataGridView1.CurrentRow != null)
112      {
113          int id = Convert.ToInt32(dataGridView1.CurrentRow.Cells[0].Value);
114
115          using (var conn = new NpgsqlConnection(connString))
116          {
117              conn.Open();
118
119              string query = @"UPDATE fields
120                             SET farm_id=@farm,
121                                 field_name=@name,
122                                 soil_type=@soil,
123                                 area=@area
124                             WHERE field_id=@id";
125
126              using (var cmd = new NpgsqlCommand(query, conn))
127              {
128                  cmd.Parameters.AddWithValue("@farm", cmbFarm.SelectedValue);
129                  cmd.Parameters.AddWithValue("@name", txtFieldName.Text);
130                  cmd.Parameters.AddWithValue("@soil", txtSoil.Text);
131                  cmd.Parameters.AddWithValue("@area", int.Parse(txtArea.Text));
132                  cmd.Parameters.AddWithValue("@id", id);
133
134                  cmd.ExecuteNonQuery();
135              }
136
137              LoadFields();
138          }
139      }
140  }
141
142  // DELETE
143  private void btnDelete_Click(object sender, EventArgs e)
144  {
145      if (dataGridView1.CurrentRow != null)

```

```

FieldForm.cs
SmartFarmingSystem
SmartFarmingSystem.FieldForm
label2_Click(object sender, EventArgs e)

141
142  // DELETE
143  private void btnDelete_Click(object sender, EventArgs e)
144  {
145      if (dataGridView1.CurrentRow != null)
146      {
147          int id = Convert.ToInt32(dataGridView1.CurrentRow.Cells[0].Value);
148
149          using (var conn = new NpgsqlConnection(connString))
150          {
151              conn.Open();
152
153              string query = "DELETE FROM fields WHERE field_id=@id";
154
155              using (var cmd = new NpgsqlCommand(query, conn))
156              {
157                  cmd.Parameters.AddWithValue("@id", id);
158                  cmd.ExecuteNonQuery();
159              }
160
161              LoadFields();
162          }
163      }
164  }
165
166  // GRID CLICK
167  private void dataGridView1_CellClick(object sender, DataGridViewCellEventArgs e)
168  {
169      if (e.RowIndex >= 0)
170      {
171          var row = dataGridView1.Rows[e.RowIndex];
172
173          cmbFarm.Text = row.Cells[1].Value.ToString();
174          txtFieldName.Text = row.Cells[2].Value.ToString();
175          txtSoil.Text = row.Cells[3].Value.ToString();
176          txtArea.Text = row.Cells[4].Value.ToString();
177      }
178  }
179
180  // BACK
181  private void btnBack_Click(object sender, EventArgs e)
182  {
183      this.Hide();
184      Dashboard d = new Dashboard();
185      d.Show();
186      this.Close();
187  }
188

```



## Statistics Form

Gui screenshots of statistics are given on DML part. Statistics form codes:

```
StatisticsForm.cs
SmartFarmingSystem
SmartFarmingSystem.StatisticsForm
btnQ4_Click(object sender, EventArgs e)

1 using Npgsql;
2 using System;
3 using System.Collections.Generic;
4 using System.ComponentModel;
5 using System.Data;
6 using System.Drawing;
7 using System.Text;
8 using System.Windows.Forms;
9
10 namespace SmartFarmingSystem
11 {
12     5 bayyuru
13     public partial class StatisticsForm : Form
14     {
15         string connString = "Host=ep-bold-surf-apre6yz3.c-7.us-east-1.aws.neon.tech;Database=neondb;Username=neondb_owner;Password=npqUsDFP10g;SSL Mode=Require;Trust Server Certificate=true;";
16         int selectedCropId = -1;
17         1 bayyuru
18         public StatisticsForm()
19         {
20             InitializeComponent();
21         }
22
23         6 bayyuru
24         void LoadData(string query)
25         {
26             using (var conn = new NpgsqlConnection(connString))
27             {
28                 conn.Open();
29                 using (var da = new NpgsqlDataAdapter(query, conn))
30                 {
31                     DataTable dt = new DataTable();
32                     da.Fill(dt);
33                     dataGridView1.DataSource = dt;
34                 }
35             }
36
37             //This query shows fields with an average temperature higher than the overall system average.
38             1 bayyuru
39             private void btnQ1_Click(object sender, EventArgs e)
40             {
41                 string q = @"
42                 SELECT f.field_name, AVG(s.temperature) AS avg_temp
43                 FROM sensorreadings s
44                 JOIN fields f ON s.field_id = f.field_id
45                 GROUP BY f.field_name
46                 HAVING AVG(s.temperature) > (
47                     SELECT AVG(temperature) FROM sensorreadings
48                 );";
49
50                 LoadData(q);
51             }
52
53             // o This query calculates the total number of crops in each farm using multiple table joins.
54             1 bayyuru
55             private void btnQ2_Click(object sender, EventArgs e)
56             {
57                 string q = @"
58                 SELECT fa.farm_name, COUNT(c.crop_id) AS total_crops
59                 FROM farms fa
60                 JOIN fields f ON fa.farm_id = f.farm_id
61                 JOIN crops c ON f.field_id = c.field_id
62                 GROUP BY fa.farm_name
63                 ORDER BY total_crops DESC";
64                 LoadData(q);
65             }
66
67             // o This query displays the minimum and maximum temperature recorded in each field.
68             1 bayyuru
69             private void btnQ3_Click(object sender, EventArgs e)
70             {
71                 string q = @"
72                 SELECT f.field_name,
73                        MAX(s.temperature) AS max_temp,
74                        MIN(s.temperature) AS min_temp
75                 FROM sensorreadings s
76                 JOIN fields f ON s.field_id = f.field_id
77                 GROUP BY f.field_name";
78                 LoadData(q);
79             }
80
81             // o This query identifies fields that frequently experience low moisture levels.
82             1 bayyuru
83             private void btnQ4_Click(object sender, EventArgs e)
84             {
85                 string q = @"
86                 SELECT f.field_name,
87                        MAX(s.moisture) AS highest_moisture
88                 FROM fields f
89                 JOIN sensorreadings s
90                 ON f.field_id = s.field_id
91                 GROUP BY f.field_name
92                 ORDER BY highest_moisture DESC
93                 LIMIT 1;";
94             }
95         }
96     }
97 }
```



```
StatisticsForm.cs x
SmartFarmingSystem SmartFarmingSystem.StatisticsForm btnQ4_Click(object sender, EventArgs e)
89 #
90 GROUP BY f.field_name
91 ORDER BY highest_moisture DESC
92 LIMIT 1;
93 ";
94 LoadData(q);
95 }
96
97
98
99 // o This query retrieves the most recent sensor reading for each field.
100 1 basipuru
101 private void btnQ5_Click(object sender, EventArgs e)
102 {
103     string q = @"
104     SELECT f.field_name, s.temperature, s.reading_date
105     FROM sensorreadings s
106     JOIN fields f ON s.field_id = f.field_id
107     WHERE s.reading_date = (
108         SELECT MAX(s2.reading_date)
109         FROM sensorreadings s2
110         WHERE s2.field_id = s.field_id
111     )";
112     LoadData(q);
113 }
114
115 // o This query lists fields that currently have no crops assigned.
116 1 basipuru
117 private void btnQ6_Click(object sender, EventArgs e)
118 {
119     string q = @"
120     SELECT f.field_name
121     FROM fields f
122     LEFT JOIN crops c ON f.field_id = c.field_id
123     WHERE c.crop_id IS NULL";
124     LoadData(q);
125 }
126
127 1 basipuru
128 private void btnBack_Click(object sender, EventArgs e)
129 {
130     Dashboard d = new Dashboard();
131     d.Show();
132     this.Close();
133 }
```

## 6) GitHub Link:

[Strawberry farming corp.](#)